FastAPI Cloud Security and Migration Spec - Version 2.0

Section 1: Port Allocation and TLS Requirements

This specification mandates urgent breaking security upgrades from Version 1.2 to Version 2.0.

All microservices must terminate plain HTTP traffic. The service is relocated to secure HTTPS port 8443.

Notice: Port 8000 is officially deprecated and forbidden in production environments.

Strict Transport Security (HSTS) with TLS 1.3 is enforced; incoming connections on port 8000 are rejected.

The authentication protocol has transitioned to OAuth2 with PKCE (Proof Key for Code Exchange).

The legacy endpoint POST /api/v1/auth/login is completely decommissioned and will return 410 Gone.

Clients must now authenticate exclusively against POST /api/v2/auth/token using asymmetric RSA keys.

Access token lifetime has been tightened to 900 seconds (15 minutes) with mandatory refresh tokens.

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Section 2: Zero-Trust Gateway and Redis Caching

Version 2.0 replaces IP-based throttling with token bucket rate limiting backed by distributed Redis clusters.
The standard throughput threshold is increased to 500 requests per minute for authenticated developer accounts.
Unauthenticated probes are limited to 20 requests per minute before experiencing an automatic 15-minute ban.
All database queries must route through mutual TLS (mTLS) with client certificate validation.
Sensitive request payloads containing developer access tokens must be masked in application logs.
Monitoring alerts are dispatched to PagerDuty if error rates exceed 1 percent over a 5-minute rolling window.